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| **Started on** | Wednesday, 30 September 2026, 8:44 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 30 September 2026, 8:58 PM |
| **Time taken** | 14 mins 45 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int a[],int l,int r,int x){
    if(l>=r)
    return 0;
    int sum=a[l]+a[r];
    if(sum==x){
        printf("%d\n%d",a[l],a[r]);
        return 1;
    }
    if(sum<x)
    return findPair(a,l+1,r,x);
    else
    return findPair(a,l,r-1,x);
}
int main()
{
    int n,x;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++)
    scanf("%d",&a[i]);
    scanf("%d",&x);
    if(!findPair(a,0,n-1,x))
    printf("No");
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.